

Timing Analysis overview

Timing Constraint of a Flip-Flop

Setup Time: Minimum amount of time the data signal must be stable before the active clock edge.

Inside the flip-flop, the signal needs time to propagate through the internal gates (like the master latch) to ensure the correct value is "ready" to be locked in when the clock strikes.

If violated, the flip-flop may capture the wrong data or enter a state of metastability, where the output hovers between 0 and 1 for an unpredictable amount of time.

Hold time: The minimum amount of time the data signal must remain stable after the active clock edge.

Even after the clock edge occurs, the internal transistors take a moment to fully "close the gate." If the data changes too quickly, the new value might "sneak" into the latch and overwrite the data you intended to capture.

If violated: The flip-flop captures the "new" data instead of the "old" data, leading to a race condition

Clock Skew:

It is a phenomenon in synchronous digital circuit systems (such as computer systems) in which the same sourced clock signal arrives at different components at different times.

Cause : Wires , Buffers etc

Clock Jitter:

Sometimes some external sources like noise, voltage variations may cause to disrupt the natural periodicity or frequency of the clock. This deviation from the natural location of the clock is termed to be clock jitter.

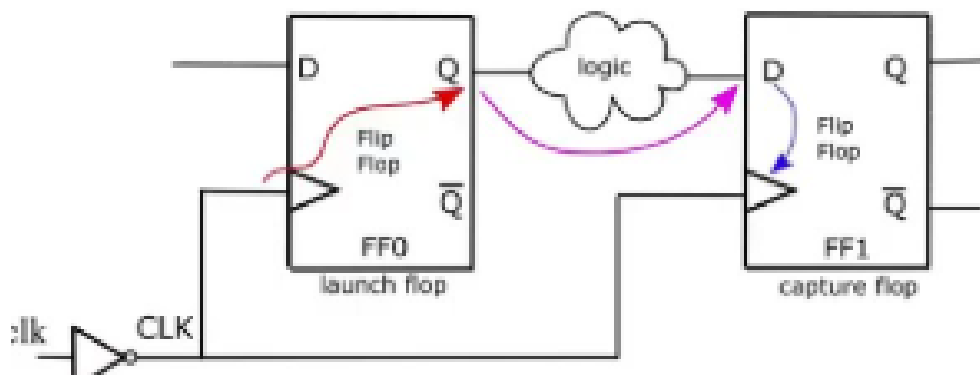
Clock Uncertainty = Clock Jitter + Clock Skew

Static Timing Analysis Concept:

Two main problems can arise in synchronous logic:

- Max Delay: The data doesn't have enough time to pass from one register to the next before the next clock edge.
- Min Delay: The data path is so short that it passes through several registers during the same clock cycle.
- Max delay violations are a result of a slow data path, including the registers, it is often called the "Setup" path.
- Min delay violations are a result of a short data path, causing the data to change before the clock has passed, therefore it is often called the "Hold" path.

Setup or Max. Constraint:



After the clock rises, it takes t_{cq} for the data to propagate to Q output of 1st FF.

- Then the data goes through the delay of the logic (t_{logic}) to get to input of next FF.
- The data has to arrive at next FF input, and the 2nd FF has a setup time of t_{setup} next clock

Neglecting clock uncertainty, Data Arrival time

$$AT = t_{cq} + t_{logic} + t_{setup}$$

Clock arrival time to next FF is: **RT = time period of clock + clock uncertainty.**

Neglecting clock uncertainty, Time period of clock **RT > AT** to meet timing.

Timing slack = Required time - Arrival time.

To meet timing, slack should be positive.

Hold problems occur due to the **logic changing before t_{hold} has passed**. This is not a function of cycle time – it is relative to a single clock edge!

For hold timing constraint, $t_{cq} + t_{logic} > t_{hold}$

Launch and Capture Path:

Launch path is launch clock path which is responsible for launching the data at launch flip flop

Capture path is the capture clock path which is responsible for capturing the data at capture flip flop.

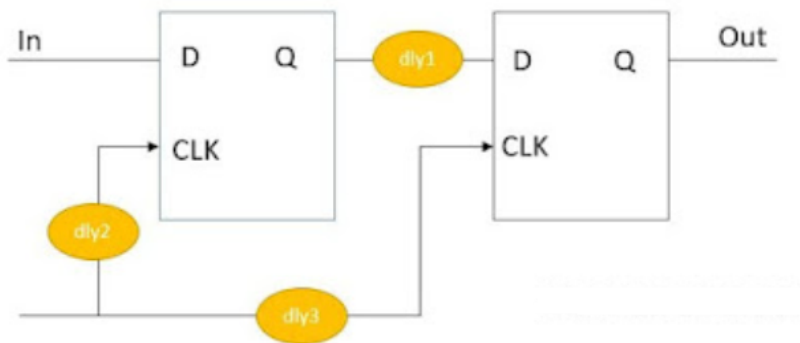
Advantages of Static Timing Analysis:

Unlike dynamic simulation, which only checks the paths exercised by specific input vectors, STA checks all possible paths in the design. It evaluates every start-to-end path, ensuring no critical path is missed, which is essential for ensuring that 100% of the timing constraints are met.

Sample problem:

Problem-1:

In the given circuit, verify whether any setup violations takes place or not



Given $dly1=5ns$, $dly2=dly3=0ns$. Time period of CLK (T_{clk}) is $9ns$. Clock to Q delay of each FF is $3ns$. Setup time of each FF is $2ns$ and hold time is $1.5ns$.

Solution:

Setup time analysis (Max delay)

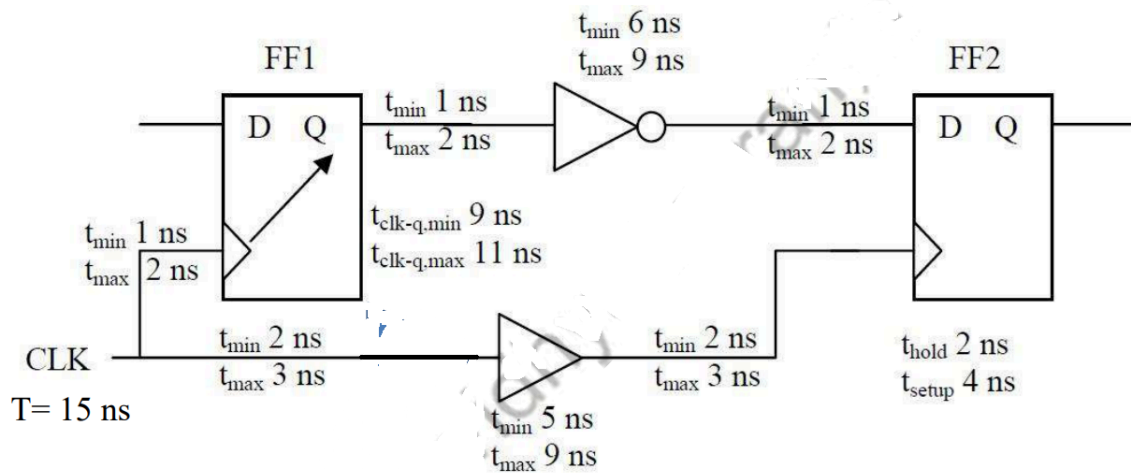
The setup time constraint when a pair of flops circuits is given will be,

$$T_{minimum} = T_{setup_time_FF2} + T_{clock_Q1} + dly1$$

$= 2ns + 3ns + 5ns = 10ns$. Since $T_{minimum} > T_{clk}$, there is setup violation.

SetupSlack is $8ns - 10ns = -2ns$, negative, hence setup violation takes place.

Problem-2:



For the circuit shown in Figure, all min and max delay is specified. Given time period of Clock is $15ns$. Find whether there is any setup or hold violations occur or not.

Solution:

Setup slack = Min. Clock Path Delay - Max. Data Arrival Time

Min. Clock Path Delay = $(15 \text{ ns} + 2 \text{ ns} + 5 \text{ ns} + 2 \text{ ns} - 4 \text{ ns}) = 20 \text{ ns}$

Note, for setup constraint, $T > T_{cq} + T_{logic} + T_{setup}$,

That gives, $T - T_{setup} > T_{cq} + T_{logic}$

That's why the setup time is subtracted.

Max. Data Arrival Time = $(2 \text{ ns} + 11 \text{ ns} + 2 \text{ ns} + 9 \text{ ns} + 2 \text{ ns}) = 26 \text{ ns}$

Setup slack = $20 \text{ ns} - 26 \text{ ns} = -6 \text{ ns}$. Hence setup violation occurs.

Hold slack calculation:

Hold time slack = Min. Data Arrival Time - Max. Clock Path Delay

Min. Data Arrival Time = $1 \text{ ns} + 9 \text{ ns} + 1 \text{ ns} + 6 \text{ ns} + 1 \text{ ns} = 18 \text{ ns}$

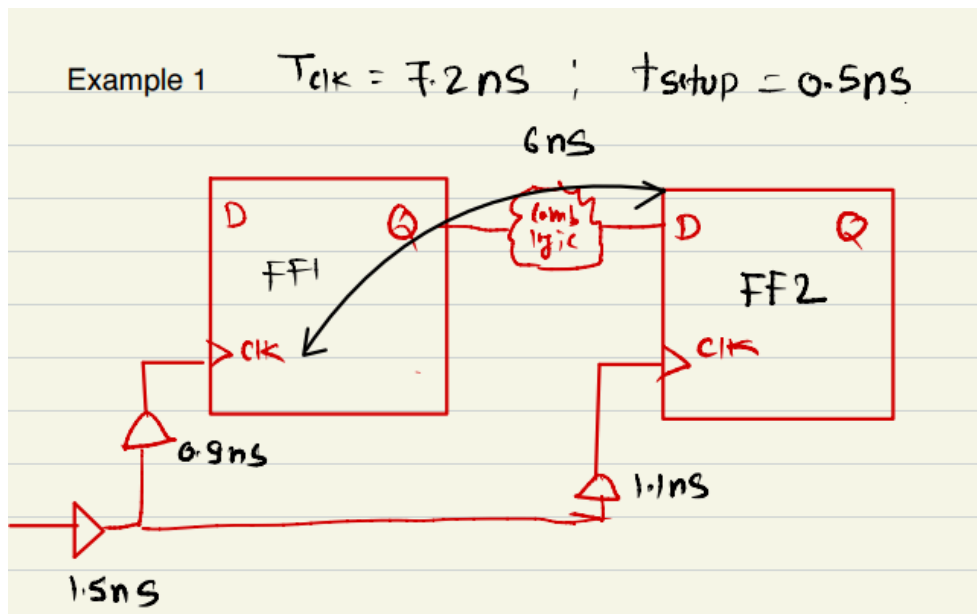
Max clock path delay = $3 \text{ ns} + 9 \text{ ns} + 3 \text{ ns} + 2 \text{ ns} = 17 \text{ ns}$

Hold slack = $18 - 17 = 1 \text{ ns}$, positive, hence no hold violation.

Note: Hold time analysis takes place at same clock cycle, that's why clock period T is not present in the calculation.

Problem-3

For the given logic circuit, verify it for any setup violation.



For the same design, if clock period $T_{clk}=5$ nsec, then find whether any violations takes place or not. What should be the max clock frequency that can be applied to this circuit without any setup timing violations?

Solution:

Data arrival time = Launch clk path + Max data path

Data Required time = Clock period + Capture clock path - $t_{setup}(ff2)$

For setup time check, Data arrival time(DAT) $\leq$ data required time (DRT)

$$\text{Launch clock path} + \text{Max Data Path} \leq \text{Clock period} + \text{Capture Clock path} - T_{\text{setup}}$$

Calculation:

Launch clock path = $1.5 + 0.9 = 2.4\text{ns}$.

Max data path = 6ns .

Capture clk path = $1.5 + 1.1 = 2.6\text{ns}$

$T_{\text{setup}} = 0.5\text{ns}$

$\text{DAT} = 2.4 + 6 = 8.4\text{ ns}$

$\text{DRT} = 7.2 + 2.6 - 0.5 = 9.3\text{ns}$

Since $\text{DAT} < \text{DRT}$, no setup violation takes place.

Case-2:

When $T_{clk}=5\text{nsec}$,

$\text{DAT} = 2.4 + 6 = 8.4\text{ ns}$

$\text{DRT} = 5 + 2.6 - 0.5 = 7.1\text{ns}$

Since $\text{DAT} > \text{DRT}$, setup violation takes place.

Min clock time period that will not cause any violation is at boundary condition i.e.

$\text{DAT} = \text{DRT}$,

$$2.4\text{ns} + 6\text{ns} = T_{clk} + 2.6\text{ns} - 0.5\text{ns}$$

That gives, $T_{clk}(\text{min}) = 6.3\text{nsec}$

Hence , max clock Freq that can be applied = $f_{\text{max}} = 1/T_{clk}(\text{min}) = 158.7\text{MHz}$.